Team Details

a. **Team name:** **FairLens AI**

b. **Team leader name:** **Shreyash Pandey**

c. **Problem Statement:** **Unbiased AI Decision**

SOLUTION SUMMARY

Brief About The Solution

CORE IDEA

What FairLens AI does

Uploads or connects datasets, measures bias, explains risky decisions, and recommends fixes before deployment.

TARGET USERS

Who uses it

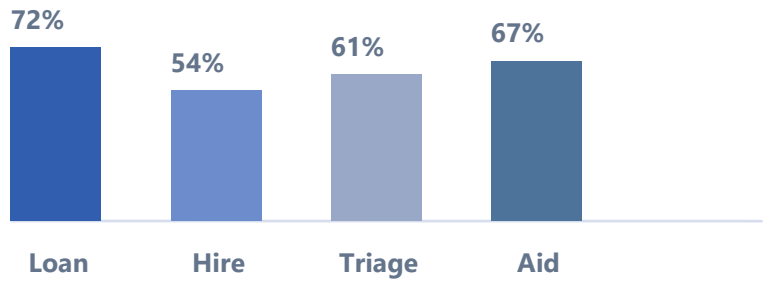
Compliance officers, data scientists, NGOs, schools, hospitals, and lending teams using automated decision systems.

GOOGLE AI

Why Google AI is included

Gemini converts fairness metrics into plain-language audit explanations and safe remediation steps.

Expected need across sectors



Prototype Value Proposition

1. Detects unfair patterns early
2. Makes fairness understandable for non-technical reviewers
3. Gives a clear audit, explain, fix workflow
4. Supports cloud deployment and future reporting

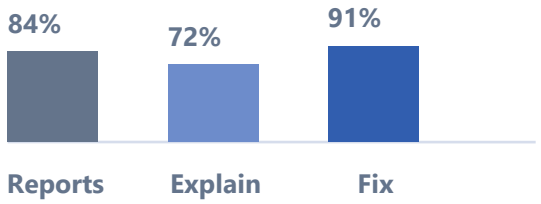
WHY THIS STANDS OUT

Opportunity, Differentiation, And USP

Different from existing ideas

Most bias tools stop at metrics. FairLens AI adds visual dashboards, plain-language explanations, and guided mitigation.

Coverage of workflow



How it solves the problem

Fairness tests identify disparity, Gemini surfaces root causes, and review steps prevent unsafe model release.

USP of the proposed solution

One platform that combines fairness metrics, decision replay, AI explanation, remediation planning, and approval tracking.

USP summary

One workflow for audit, explanation, mitigation, and approval tracking.

FEATURE MAP

Features Offered By The Solution

SCAN

Bias scan

Checks parity gaps across groups and flags imbalance quickly.

INSIGHT

Gemini narratives

Explains fairness risks in a way judges and reviewers can understand.

TRACE

Decision replay

Visualizes the path of one model decision and highlights risky inputs.

FIX

Mitigation planner

Suggests thresholds, reweighting, and human review checkpoints.

REPORT

Audit reports

Creates shareable fairness summaries with action items.

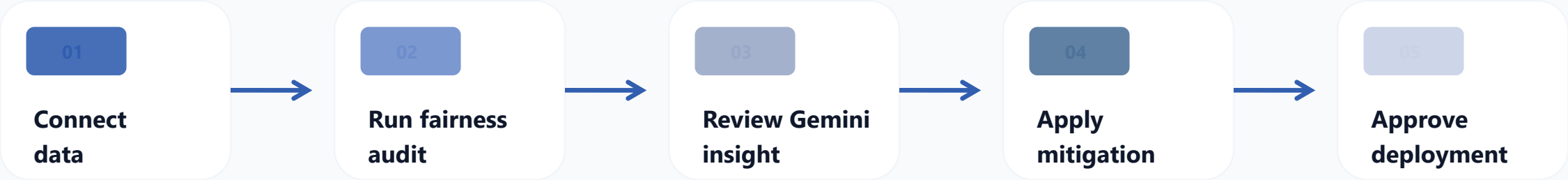
WATCH

Live monitoring

Tracks post-deployment drift so fairness does not degrade silently.

END-TO-END USER JOURNEY

Process Flow Diagram

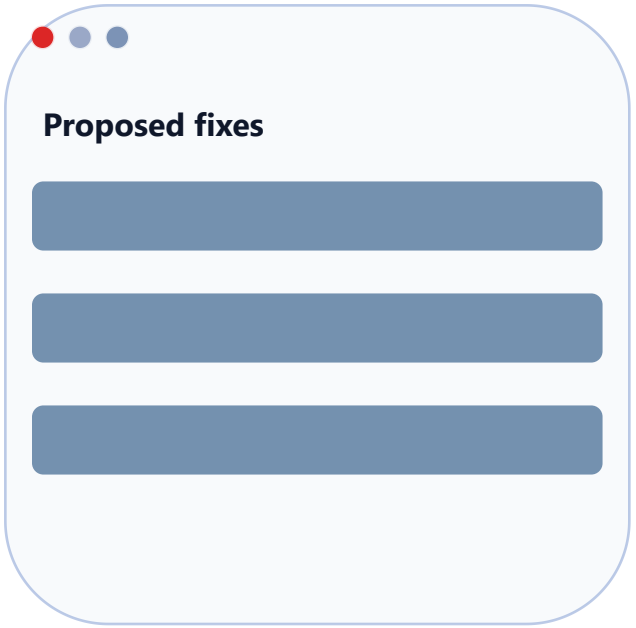


VISUAL PROTOTYPE

Wireframes / Mock Diagrams

Admin fairness dashboard

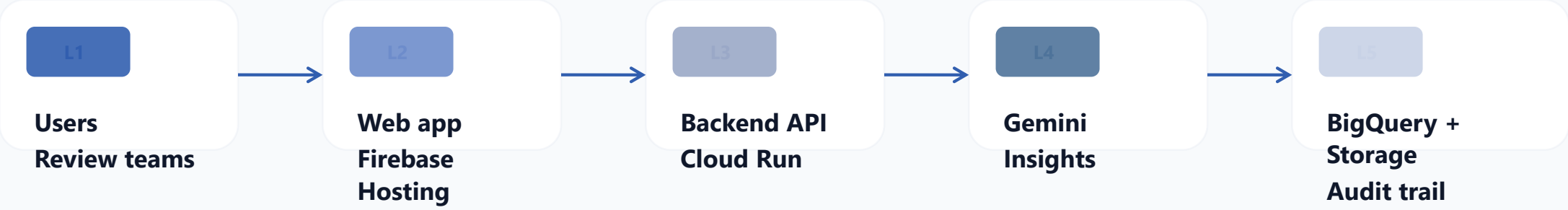
Decision replay panel



Mitigation sandbox

CLOUD + GOOGLE AI

Architecture Diagram



Why this architecture is strong

The user reviews results in the web app, Cloud Run handles processing, Gemini explains risky decisions, and BigQuery stores long-term audit history.

STACK OVERVIEW

Technologies Used In The Solution

UI

Frontend Layer

HTML, CSS, JavaScript, responsive dashboard design, and simple chart components for an accessible review experience.

INTELLIGENCE

AI + Analytics Layer

Gemini API, fairness scoring logic, explainability prompts, bias-gap calculations, and remediation recommendations.

DEPLOYMENT

Cloud Layer

Firebase Hosting, Cloud Run, BigQuery, Cloud Storage, and deployment-ready workflows for scaling the MVP.

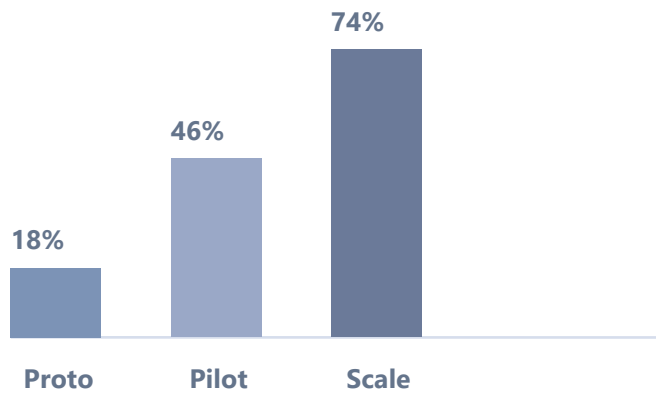
Toolchain Fit

- Fast to prototype
- Easy to demo
- Cloud-native
- Judge-friendly
- Scalable later

COST OVERVIEW

Estimated Implementation Cost

Relative cost growth



LOW

Prototype

Mostly free-tier usage with lightweight Gemini requests and static hosting.

MEDIUM

Pilot

Moderate spend for inference, storage, and audit history as teams start testing.

HIGHER

Scale

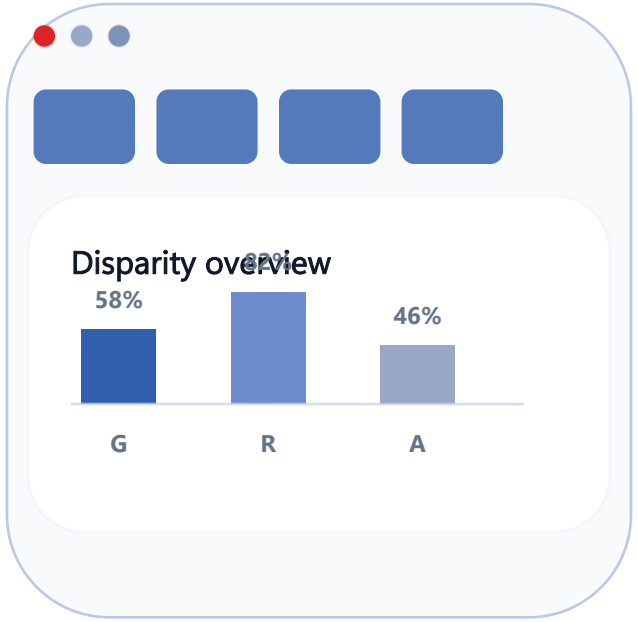
Higher cloud usage but recoverable through subscriptions for regulated sectors.

Cost strategy

Start lean, validate with pilots, then scale into a subscription-based fairness auditing platform for institutions.

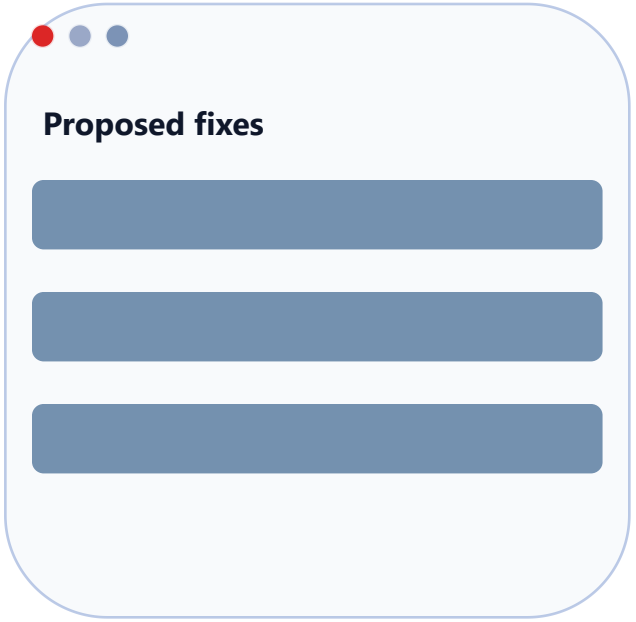
PRODUCT PREVIEW

Snapshots Of The MVP



Audit dashboard

Gemini explanation



Mitigation planner

ROADMAP

Additional Details / Future Development

01

Sector templates

Hiring, lending, healthcare, and education specific audit templates.

02

Policy packs

Localized fairness checks, documentation, and report formats.

03

Live monitoring

Production drift alerts and recurring fairness checks after deployment.

04

Approval workflow

Human sign-off, remediation history, and stronger audit trails.

Future vision

The long-term product can become a fairness operating system for institutions that use AI in high-impact decisions.

FINAL CHECKLIST

Submission Links

GitHub Public Repository

https://github.com/myself-archita/fairlens-ai

Demo Video Link (3 minutes)

https://drive.google.com/file/d/1WzmRcHT3WN2GQS69DkW_YuUjTL90YrnS/view?usp=drive_link

MVP / Working Prototype Link

https://fairlens-ai.vercel.app

WHY THIS IS SUBMISSION-READY

Impact And Readiness

Social impact

High

Targets unfair outcomes in sectors that affect livelihoods and opportunities.

Deployment readiness

Cloud-ready

Designed for Firebase Hosting, Cloud Run, Gemini, and audit data storage.

Judge clarity

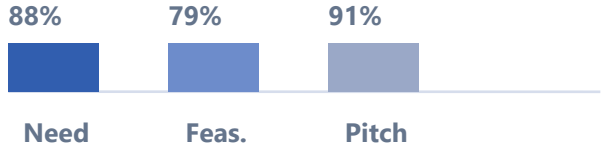
Visual

Easy to explain with charts, flows, screenshots, and practical use cases.

Why this prototype is strong

- **Solves a clear real-world problem**
- **Includes a Google AI model in the workflow**
- **Has a feasible deployment path, not just an abstract idea**

Submission strength



FAIRLENS AI

Thank You

Building fairer automated decisions with explainable AI.

FairLens AI combines bias detection, Gemini-powered explanation, remediation planning, and deployment-ready architecture in one student-friendly prototype.

Bias Detection

Gemini Insights

Cloud Ready